

Engineered Observation Limits: Secret Sharing, Leakage Channels, and Two Entropy Accountings in Exact Finite Models

Andrew H. Bond*

San José State University, San José, California, USA

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Cryptography is the engineering discipline of declared observation limits. Where the rest of this campaign has to find the consumer's channel inside a physical model, cryptography writes the channel down on purpose and then argues about what a consumer can read through it. This paper carries the campaign's consumer-relative audit to that argument in four exact finite models and to nothing else. Every model here is a declared toy model small enough to enumerate. Nothing in this paper bears on computational hardness or on asymptotic security, no real cipher, protocol, implementation, or deployed system is modeled or evaluated, and no attack is developed. What is measured is evaluation methodology, meaning the reasoning that carries a leakage measurement to a conclusion about what a reader can do. A threshold secret-sharing scheme over a field of thirty-one elements, enumerated by integer counting over all 29791 outcomes, gives an information curve that is a step, exactly zero at every subset of shares below the declared threshold with the reader's posterior exactly the prior, and exactly the whole secret at every subset above it, with no subset of any size anywhere in between. The campaign's naturally occurring redundancy plateau rose smoothly from 0.8118 to 1.9996 bits instead, so a declared threshold and a physical one differ in shape and the difference is that one of them was designed. A declared control with short randomness carries 2.6323 bits to every single reader, so the zeros belong to the scheme rather than to the measurement, and a drafted control that publishes a redundant copy of a share carries exactly nothing, which is why the bar moved to the short-randomness control before the run rather than after it. Declared leakage channels are experiments in Blackwell's sense, and a strictly positive conditional information proves non-garbling with no tolerance and no search, which is the transferable part of this work. The channel that reads a Hamming weight carries twice the information of the channel that reads one bit and is still not above it in Blackwell's sense, the two conditional informations measuring 1.8113 and 0.7806 bits. In two thousand declared random channel pairs, 268 pairs were ordered unanimously by all seven declared divergences while proved incomparable, and every one of the 268 carried a declared task that strictly prefers the channel every divergence ranks as leakier, with no such reversal anywhere among the 867 comparable pairs. The measured statement is that a reduction in a scalar leakage measure is not a guarantee for a reader's task outside certified comparability, and its constructive half is that the conditional information certificate decides comparability by proof, so an evaluation that establishes comparability may rely on its scalar ordering. Two consumers price one declared source at 3.477 bits and at exactly 1 bit, both correct, the disagreement lying entirely in the task. One run failed, on a declared clause requiring the difference of the two accountings to increase along a ladder of sources. It measured 0, 0.398, 1.337, 2.275, 2.635, 2.113, and 0.599 bits and turned over, because both accountings vanish together as a source becomes certain. The corrected registration measures the ratio, which falls strictly where the difference does not, and locates the interior maximum by bisection at spike weight 0.35338 with a difference of 2.6398 bits.

I. INTRODUCTION

This campaign models observation as consumption. A consumer reads a hidden state through a declared channel, spends a declared budget on the read, and acts on a declared task, and every claim is settled by exact measurement in a declared finite model against bars fixed before the run [9, 10]. Cryptography is the one engineering discipline that already works this way. Elsewhere the campaign has to find the consumer's channel inside a physical model and argue that the choice was not made to suit the answer. Cryptography writes the channel down on purpose, in a specification, before anyone measures

anything.

Three of its central objects are campaign objects under other names. Semantic security is restricted distinguishability, meaning a statement that a reader with declared powers cannot tell two things apart [4]. A side-channel adversary is a consumer with a declared channel, which is what the campaign calls a restricted observer [7]. Threshold secret sharing is an engineered redundancy plateau, built by hand where the campaign's quantum track found one in an environment [2, 10]. Shannon's information-theoretic account of secrecy is the oldest statement in the campaign's own language, a statement about what a reader can recover from what a reader can see [1].

The limit of what can be said here is severe and is stated once here, once in the abstract, and once in the ledger of Sec. VI. This paper measures exact finite models, so it can say nothing about computational hardness, nothing

* andrew.bond@sjsu.edu; ORCID: 0009-0003-2599-6158

about asymptotic security, and nothing about any concrete algorithm or deployment. Every model is a declared toy model. No real cipher, protocol, implementation, or deployed system is modeled or evaluated. No attack is developed and no attack is evaluated. What remains is evaluation methodology, meaning the reasoning that carries a leakage measurement to a conclusion about what a reader can do, and that reasoning can be audited exactly in settings where every alternative reading is computable rather than arguable.

Three claim types run through the paper and are never conflated. Instrument claims state that leakage informativeness, garbling relations between declared channels, and optimal reads can be computed exactly in finite models, and they are checked against closed forms. Structural claims state that a named evaluation statement is channel-relative or measurably is not, inside a declared model with every alternative channel enumerated, and they carry the campaign's exploratory label. Claims about real systems are absent. No run in this paper is labeled as supporting one, and the limit of the previous paragraph is the reason. A reader looking for evidence about the strength of a deployed system will find none here.

An anti-circularity contract governs every run. No leakage model was chosen after inspecting which choice favors a wanted verdict. No adversary task was selected after seeing the answer. No divergence or entropy measure was added to a battery after seeing the battery's verdict. Channels, tasks, budgets, priors, ladders, seeds, and bars were written into the track document before the runs they govern. One bar was moved before its run, because the object it was drafted against provably could not fail it, and the move and its reasoning were recorded in the track document before the run rather than afterward. Sec. II reports what that object measured.

The paper reports four runs and one correction. The first is the instrument layer, an engineered redundancy plateau computed by exact integer counting and validated against the known answer for threshold secret sharing. The second orders declared leakage channels by Blackwell's relation and replaces a search with a proof. The third measures where a scalar leakage ordering and a task ordering disagree. The fourth prices one declared source for two consumers with different tasks. The fourth run failed its declared bar on one clause, the failure is reported with the design error named, and the corrected registration is reported beside it.

II. AN ENGINEERED REDUNDANCY PLATEAU IS A STEP AND A NATURAL ONE IS A RAMP

The substrate is Shamir's threshold scheme over the prime field with 31 elements, with threshold 3 and 5 shares [2]. A secret uniform on the field and two uniform independent coefficients determine a share vector at the evaluation points 1 through 5, so every distribution in

this section is a function of $31 \times 961 = 29791$ equally weighted outcomes. Every distribution was computed by enumerating all of them with integer counting and never by sampling. Informations are built from integer count ratios, so a ratio that is exactly one contributes exactly zero, and every summation over outcomes uses compensated addition. The declared tolerance is 10^{-12} wherever a tolerance is needed at all.

The measured information curve as a function of the number of shares read is

$$(0, 0, \log_2 31, \log_2 31, \log_2 31) \text{ bits}, \quad (1)$$

with $\log_2 31 = 4.954196310386875$. All 15 subsets below the declared threshold carry mutual information exactly 0.0 bits rather than merely inside the tolerance, and for every share value of positive probability the reader's posterior on the secret is the uniform prior with deviation exactly 0.0 at every field element. All 10 threshold subsets carry exactly $\log_2 31$ bits with conditional entropy exactly 0.0, and an independent route, Lagrange interpolation at zero over the field, recovered the secret in all 297910 subset-outcome checks with zero failures. The exactness is structural rather than fortunate. Below the threshold every count ratio in the sum is the integer one, and at the threshold every posterior is a point mass. The step is a step in the strongest available sense, the maximum distance from any of the 31 nonempty subsets to the nearer of the two plateau values being exactly 0.0, so no subset of any size takes an intermediate value. Mutual information computed from the joint distribution and mutual information computed as a difference of entropies agree across every subset tested in all three schemes of this section to 1.8×10^{-15} .

Figure 1 places that curve beside the campaign's naturally occurring plateau. In the quantum track a system interacts with an environment and an observer reads a fragment of the environment, and the information a fragment carries rose smoothly through 0.8118, 0.9571, 0.9998, 1.0426, 1.1878, and 1.9996 bits as the fragment grew from one environment qubit to six [10]. The height of that plateau and the speed of the approach to it are set by how strong the records the interaction wrote are. The engineered plateau has no approach at all. It is a step, and it stands exactly where the designer put the threshold. The comparison is between two declared models and is not a statement about physical systems.

The declared negative control decides whether the zeros belong to the scheme or to the measurement. The control is a short-randomness scheme, a degree-one polynomial whose single coefficient is uniform on the declared five-element set rather than on the field, with shares at the same five evaluation points. Its declared bar required at least one single-share reader to carry more than 0.1 bits. Every one of its five single-share readers carries

$$\log_2 \frac{31}{5} = 2.6322682154995127 \text{ bits}, \quad (2)$$

matching the closed form with deviation exactly 0.0 and

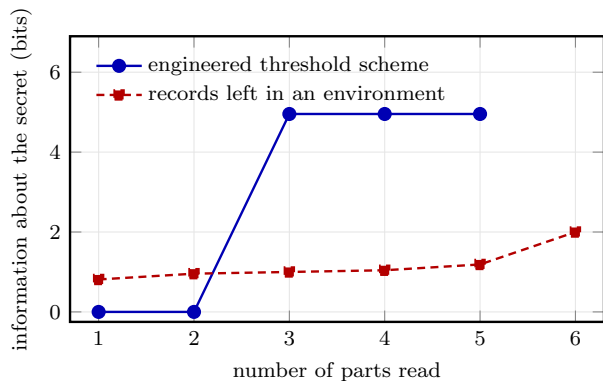


FIG. 1. Two redundancy plateaus. The engineered curve is the mutual information between the secret and a subset of shares in the declared threshold scheme, exactly zero at one and two shares and exactly $\log_2 31$ bits at three, four, and five, with no subset of any size taking an intermediate value. The other curve is the campaign’s earlier measurement of how much a fragment of an environment carries about a system after a declared interaction has written records into it, rising smoothly from 0.8118 to 1.9996 bits with the number of environment qubits read. Both curves are read from committed evidence records.

clearing the declared bar by a factor of 26, and its two-share readers all sit at $\log_2 31$, the same step one rung earlier. The instrument sees leakage when leakage exists.

The drafted control is the recorded deviation and the section’s second finding. It was a degree-one polynomial with a field-uniform coefficient whose five shares are published together with the redundant value formed by adding the secret to the coefficient. That scheme cannot leak to any single reader, because every one of its six published values is the secret masked by an independent field-uniform quantity and is therefore exactly uniform and exactly independent of the secret, and its sixth value is bit for bit the first share. A control that provably cannot fail its bar is not a control, so the bar moved to the short-randomness scheme before the run, and the drafted scheme was measured without a bar with its outcome predicted in advance so that the run could refute the prediction. It did not. All six single-share informations came out exactly 0.0, the sixth published value was identical to the first share at every one of its 961 outcomes, and exactly one of the 15 pairs of published values is uninformative, the pair of identical values, with the other 14 determining the secret. Publishing a redundant copy of a share leaks nothing to any single reader. Redundancy and leakage are different things, which the campaign’s quantum track reached from the other side when it found that redundant records are what make a state readable by many observers at once [10].

TABLE I. The five declared leakage channels, what each carries about the declared key, and the certificate that decides its position under the identity channel. The conditional information of key and channel given the identity reading is exactly zero for all four non-identity channels, which proves each is a garbling of the identity. All values are from the committed evidence record.

Channel	information (bits)	conditional information given the identity (bits)
identity	4	
Hamming weight	2.0306	0
low bit	1	0
noisy Hamming wt.	0.8867	0
constant	0	0

III. A CONDITIONAL INFORMATION CERTIFICATE PROVES NON-GARBLING WITHOUT A THRESHOLD

A declared key-dependent leakage model is an experiment in Blackwell’s sense, a row-stochastic matrix whose rows are the reading distributions in each hidden state [3]. Blackwell’s theorem says that one experiment is at least as good as another for every prior and every task exactly when the second is a garbling of the first, meaning the second can be produced from the first by post-processing that does not see the hidden state. The question of whether one declared side channel is better than another therefore has a precise answer with two possible forms. Either one garbles the other, and then the ordering holds for every task a reader might have, or neither garbles the other, and then no single number can order them for every task.

The substrate is a four-bit key uniform on sixteen values, a declared bijective substitution table, and five declared channels that read the substituted value. The identity channel reads the value. The Hamming-weight channel reads the number of set bits. The low-bit channel reads the lowest bit. The noisy Hamming-weight channel adds declared symmetric noise of probability 0.2 in each direction to the Hamming weight. The constant channel reads nothing. Table I gives what each carries about the key.

The certificate is the transferable part of this run, and it replaces a search with a proof. Channel B is a garbling of channel A exactly when B is conditionally independent of the key K given A , so

$$I(K; B | A) = 0 \iff B \text{ is a garbling of } A, \quad (3)$$

and a strictly positive left-hand side proves that B is not a garbling of A with no tolerance and no search involved. The projected-gradient garbling search used elsewhere in the campaign was retained here only as a cross-check. An earlier run in the campaign established incomparability by thresholding that search residual, the threshold was chosen without computing the scale it would be compared against, the run failed its own bar, and a separate registration had to repair it with an analytic bound [9]. Equation (3) removes the threshold from the argument entirely.

All five declared items passed. The identity channel is above every declared channel, each search residual exactly zero and each conditional information given the identity exactly zero. The constant channel is below every declared channel, each residual exactly zero as well. Adding the declared noise to the Hamming weight is a garbling, residual 2.2×10^{-16} and conditional information 4.8×10^{-17} , and the reverse direction is proved impossible without any tolerance argument, the conditional information of key and Hamming weight given the noisy reading measuring 1.1439 bits, which is exactly the difference between what the two channels carry. The data-processing control holds across the declared battery of 201 key-recovery tasks, 200 seeded random utilities on sixteen actions plus the maximum-likelihood key-guessing task, with a worst task gain under garbling of 5.6×10^{-17} .

The pair that matters is the incomparable one. The Hamming-weight channel carries 2.0306 bits about the key and the low-bit channel carries exactly 1, so a scalar comparison ranks the first above the second by better than a factor of two. Yet 1.8113 bits of the Hamming weight survive knowing the low bit and 0.7806 bits of the low bit survive knowing the Hamming weight, so neither is a garbling of the other and no scalar comparison orders them for every task. Carrying twice as much information and being better for every task are different properties, and in this declared model the first holds while the second fails. The next section measures how often the two part company and what happens to a reader's task when they do.

IV. UNANIMOUS SCALAR LEAKAGE ORDERINGS CARRY NO TASK GUARANTEE OUTSIDE CERTIFIED COMPARABILITY

A single declared secret bit, uniform, is read through channels into a declared four-symbol alphabet, so every channel is a two by four row-stochastic matrix and every scalar fidelity measure between the two conditional reading distributions is an f -divergence. The declared battery holds seven, the two Kullback-Leibler directions, total variation, squared Hellinger, the two chi-squared directions, and Jensen-Shannon [6]. The declared task battery holds 205, two hundred seeded random utilities on three actions and five commitment tasks in which a reader must guess the secret bit and pays a declared cost of 2, 5, 10, 20, or 50 for being wrong. A task reversal is a pair of channels in which all seven divergences strictly order one channel above the other and some declared task strictly prefers the other.

The declared exhibit contrasts broad mild separation against a rare decisive symbol. Channel A has rows (0.55, 0.25, 0.15, 0.05) and (0.45, 0.25, 0.15, 0.15), so its two reading distributions differ a little on most symbols. Channel B has rows (0.49, 0.49, 0.01, 0.01) and (0.49, 0.49, 0.005, 0.015), so it is nearly blind except on a rare symbol that is three times more likely under one

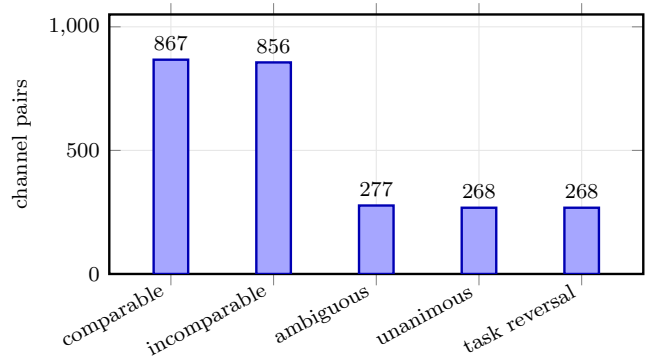


FIG. 2. The declared ensemble of 2000 random leakage channel pairs. The first three bars partition the ensemble by certificate, comparable when a garbling is certified in some direction, incomparable when both searches are refused above the declared 0.01, ambiguous otherwise and excluded as declared. The last two bars are subsets of the incomparable class, pairs ordered unanimously by all seven declared divergences and pairs whose unanimous ordering is reversed by a task in the declared battery. The two subsets are equal at 268. No comparable pair carries a reversal. Every count is read from the committed evidence record.

value of the secret. The reasoning was declared before the run so that the run could refute it, and the run confirmed it. All seven divergences order A above B , the smallest margin being 0.0149, while a declared commitment task prefers B by 0.00212 because the rare decisive symbol is the only thing a reader who must commit can act on. The pair is proved incomparable in both directions, the conic lower bound on any garbling residual measuring 0.00175 in one direction and the total-variation difference measuring 0.095 in the other, a garbling being unable to raise an f -divergence. Neither of those two numbers is a search residual.

Two further items were declared with the exhibit. The data-processing control drew five hundred declared garbled pairs and checked them against the full task battery. No divergence rose under any garble, the smallest drop in the required direction measuring 0.00262, and no task value rose by more than 3.3×10^{-16} , which is Blackwell's theorem and the data-processing inequality holding inside the instrument. A violation there would have indicted the instrument rather than the theorems. The localization item classified a declared ensemble of 2000 seeded random channel pairs by certificate and required every task reversal to fall in the incomparable class. Figure 2 shows the outcome, 867 comparable, 856 incomparable, 277 ambiguous, and zero task reversals among the comparable pairs, with 210 among the excluded ambiguous pairs. The prevalence measurement was declared with no bar. Among the 856 certified-incomparable pairs, 268 are ordered unanimously by all seven declared divergences, and every one of those 268 carries a task reversal in the declared battery, the largest reversal margin being 0.0796 and the median 0.0183. The campaign's decision-theoretic

track measured the same structure in a different substrate and found 281 of 281 [9].

The reading was fixed in the track document before the run and is binding on this report. A reduction in a scalar leakage measure is not a guarantee for a reader's task outside certified comparability, because in every unanimously ordered incomparable pair measured here the ordering the scalar measures report and the ordering a task reports disagree. This is a statement about evaluation methodology in declared finite models. It is not a claim that any countermeasure, implementation, or system is insecure, and none was modeled. The constructive half of the statement matters as much and is measured by the same runs. Equation (3) decides comparability by proof rather than by search, comparability is common rather than rare in this ensemble at 867 of 2000 pairs, and Blackwell's theorem held exactly where it must, since not one comparable pair carried a reversal. An evaluation that first establishes comparability may therefore rely on its scalar ordering for every task, and the certificate is cheap enough to compute for any pair of declared finite channels.

V. THE DIFFERENCE BETWEEN TWO ENTROPY ACCOUNTINGS TURNS OVER AND THEIR RATIO DOES NOT

The same declared source is read by two consumers. One averages over many independent reads and is charged the Shannon entropy. One makes a single guess and is charged the min-entropy. Neither accounting is a property of the source alone. This run measures where they diverge, and it is the run that failed a declared bar.

Two declared families of sources on an alphabet of 32 outcomes carry every probability as an exact rational. Family U places $1/32 + \epsilon$ on outcome zero and spreads the rest uniformly, with ϵ on the declared ladder 0, 0.01, 0.05, 0.15, 0.30, 0.60, 0.90, so the first member is exactly uniform and the last is a hard spike. Family S places q on outcome zero and spreads the rest uniformly, with q on the declared ladder 0.5, 0.9, 0.99. The declared key-material case is the member of S at $q = 0.5$, half the mass on one outcome and half spread over the remaining 31. Every distribution is built from exact rationals so the probabilities sum to exactly one and the code construction compares rationals rather than rounded floats.

Three instrument items passed sharply and were unchanged by the correction. Summed Shannon entropy agrees with its closed form at every one of the ten declared sources to 8.9×10^{-16} , and the min-entropy computed from the exact numerator and denominator of the largest probability agrees with the direct route to the same 8.9×10^{-16} , with the largest probability located by exhaustive scan over the alphabet rather than assumed. The optimal single-guess success probability, found by exhaustive maximization over the alphabet, equals two to the power minus the min-entropy with worst devia-

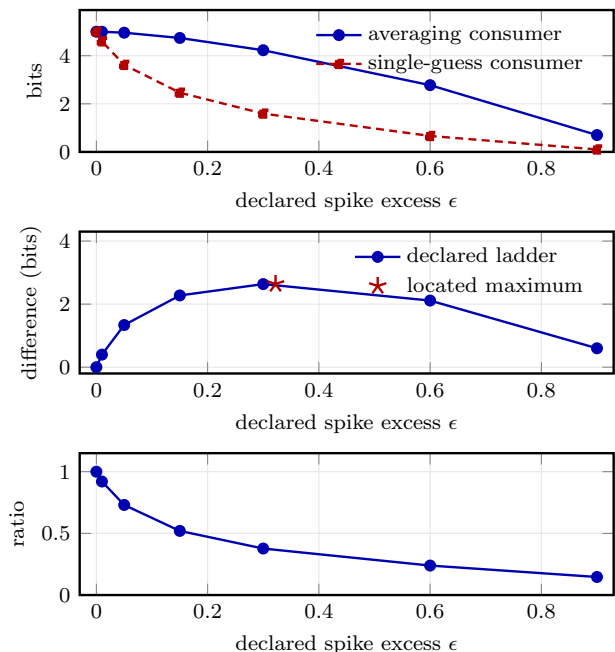


FIG. 3. Two accountings of one declared source family along the declared ladder. The top panel shows what the averaging consumer and the single-guess consumer are charged. The middle panel shows the number of bits separating them, which rises for four ladder steps and falls for two, with the star marking the interior maximum located by bisection on the closed-form derivative at spike weight 0.35338, where the separation reaches 2.6398 bits. The declared ladder came within 0.0045 bits of that maximum and stepped over it. The bottom panel shows the fraction of the averaging consumer's accounting that survives to the single-guess consumer, which falls strictly at every ladder step. All values are read from committed evidence records.

tion 1.4×10^{-17} , and eight of the ten sources are bit for bit identical. A Huffman code was constructed for every source with a heap and no library import [5], every code satisfies the Kraft equality exactly, and every mean code length sits at or above the Shannon entropy and strictly below the entropy plus one, the classic bound [6]. The smallest lower margin is exactly 0.0 at the uniform member, where the mean length is exactly 5 bits and the entropy is exactly 5 bits, and the smallest upper margin is 0.0807 bits at $q = 0.99$, where the mean length 1.0497 sits below the entropy plus one of 1.1303.

The failed item is the fourth, and the failure is in one clause of the declaration rather than in the substrate. The clause declared that the min-entropy is at most the Shannon entropy at every source, that the two agree exactly at the uniform member and nowhere else, and that the difference between them increases strictly along the ladder. The first two held, the worst value of min-entropy minus Shannon entropy measuring exactly 0.0 and the equality holding exactly at the uniform member. The third failed. The measured difference along the ladder

is

$$0, 0.398, 1.337, 2.275, 2.635, 2.113, 0.599 \text{ bits}, \quad (4)$$

so the first four increments are positive and the last two are negative, and the recorded verdict of the run is FAIL.

The design error is named in the record. The declaration took a fact about the near-uniform end of the ladder and extended it to the whole ladder without checking the other end. Both accountings go to zero together as the spike takes all the mass, because a source that is almost certain is cheap to a consumer who averages and cheap to a consumer who guesses once, so the difference has to turn around and its maximum is interior. The declared ladder crossed that maximum between $\epsilon = 0.30$ and $\epsilon = 0.60$. The difference of the two accountings was the wrong quantity to demand monotonicity from.

The correction replaced the refuted clause with a monotone statement the substrate supports and added the location of the interior maximum as a measurement, changing nothing else. The ratio of the two accountings falls strictly along the whole ϵ ladder, through 1.0, 0.9203, 0.7304, 0.5199, 0.3769, 0.2390, 0.1464, and strictly along the whole q ladder, through 0.2876, 0.1576, 0.1112. The closed-form derivative of the difference with respect to the spike weight a is

$$\frac{d}{da} [H(a) - H_\infty(a)] = \log_2 \frac{1-a}{31a} + \frac{1}{a \ln 2}, \quad (5)$$

which reads 46.166, 34.559, 16.301, 5.181, 0.4147, -3.444 , and -7.165 along the ladder, one sign change, falling inside the bracket declared for it before the run. That derivative was checked against a central finite difference of the measured difference at every ladder point and agreed to 1.6×10^{-8} against the declared 10^{-6} , so the unimodality statement rests on a derivative that was checked against the quantity it differentiates. Bisection inside the declared bracket converged in 39 steps to a width of 5.5×10^{-13} and places the maximizing spike weight at 0.35338, where the separation reaches 2.6398 bits and the derivative reads 4.5×10^{-12} . The largest separation on the declared ladder was 2.6353 bits, so the ladder came within 0.0045 bits of the true maximum and stepped over it. The corrected registration passed all three of its items. Figure 3 shows all of it.

The measurement that carried no bar in either registration is the key-material case. For a source with half its mass on one outcome and half spread uniformly over the remaining 31, the averaging consumer is charged 3.477098155193437 bits and the single-guess consumer is charged exactly 1 bit, a factor of 3.48 between the two prices. The single guess succeeds with probability exactly 0.5 and a Huffman code for the same source averages 3.4839 bits per symbol. Both numbers are correct readings of one substrate and the disagreement is entirely in the task. Which entropy is the right entropy is set by the consumer's task and not by the source. Nothing in this section is a claim about any real key-generation system, and no such system is modeled.

VI. WHAT THIS PAPER ESTABLISHES

Table II registers the runs with their recorded verdicts. Read together they answer the arc's question inside its declared scope, which of the security statements available in these models are bookkeeping relative to a declared consumer and which are properties of the substrate no declared consumer can move.

Three findings are consumer-relative in the precise sense that a declared change moves them with the substrate held fixed. Which of two leakage channels is better depends on the reader's task whenever the two are incomparable, and incomparability is not rare, since 856 of 2000 declared random pairs were certified incomparable and 268 of them were ordered unanimously by every scalar measure in the battery while a declared task reversed that order. Which entropy prices a source depends on the consumer's task, and the same declared source costs 3.477 bits to a consumer who averages and exactly 1 bit to a consumer who guesses once. How a redundancy plateau approaches its height depends on whether a designer placed it, since the engineered plateau is a step with nothing in between while the plateau the quantum track found rose smoothly with the strength of the records that produced it.

Two findings are the opposite, properties no declared consumer moves. Below the declared threshold the secret-sharing scheme gives every reader exactly the prior, and that zero is exact rather than small, so no choice of task, budget, or reading strategy reaches the secret. And a garbling never helps any task, which held across every declared control here at the level of 10^{-16} , so wherever comparability is certified a scalar ordering is a task ordering. The constructive consequence is stated with the negative one and measured by the same runs. The certificate of Eq. (3) decides comparability by proof with no threshold, and an evaluation that establishes comparability may rely on its scalar ordering.

The run that failed is part of the result. A declared clause demanded that the number of bits separating two accountings increase along a ladder of sources, the ladder was long enough to reach the far end, and the difference turned over because both accountings vanish together on a source that is almost certain. The design error was in the choice of quantity rather than in the substrate or the instrument, and the corrected registration measures the ratio, which is monotone, and locates the maximum of the difference, which is interior. A campaign that computes its verdicts from declared items records that sequence rather than the endpoint.

These are structural results about declared finite models carrying the campaign's exploratory label. They are about evaluation methodology. They do not show that any real cipher, protocol, implementation, or deployed system is strong or weak, no experiment reported here can bear on that question, and no attack was developed or evaluated. A reader who wants to carry anything from this paper into practice can carry one thing, which is that the certificate is computable and settles by proof whether a scalar leakage

TABLE II. The ledger of the crypto arc. Verdicts are recorded exactly as computed from the declared items of each run, including the failure. Measured in model, exploratory label, means an exact measurement in a declared finite construction with append-only evidence records and bars, seeds, ladders, and batteries fixed before the run. Key numbers are quoted from the committed records. The last row is the scope of the whole arc and is a non-claim.

Run	Verdict	Key measurements	Label
CR-0 secret-sharing plateau	PASS	information curve exactly $(0, 0, \log_2 31, \log_2 31, \log_2 31)$ bits over 29791 enumerated outcomes, all 15 below-threshold subsets exactly 0 with posterior deviation exactly 0, all 10 threshold subsets exactly $\log_2 31$ with conditional entropy exactly 0 and Lagrange recovery in 297910 of 297910 checks, maximum distance to the nearer plateau exactly 0, short-randomness control at $\log_2(31/5) = 2.6323$ bits per single share against a 0.1 bit bar, redundant-share control exactly 0 to every single reader with 14 of 15 pairs determining, route agreement 1.8×10^{-15}	instrument
CR-1 leakage ordering	PASS	channel informations 4, 2.0306, 1, 0.8867, and 0 bits, every non-identity channel a garbling of the identity with conditional information exactly 0, declared noise a garbling at residual 2.2×10^{-16} and conditional information 4.8×10^{-17} with the reverse proved impossible at 1.1439 bits, Hamming weight and low bit proved incomparable at 1.8113 and 0.7806 bits, data-processing control over 201 tasks with worst gain 5.6×10^{-17}	instrument, and measured in model, exploratory
CR-2 scalar orderings against tasks	PASS	exhibit proved incomparable, conic bound 0.00175 and total-variation difference 0.095, all seven divergences ordering A above B by at least 0.0149 while a declared commitment task prefers B by 0.00212, garbling control over 500 pairs with worst task gain 3.3×10^{-16} , ensemble of 2000 pairs giving 867 comparable, 856 incomparable, 277 ambiguous, 268 unanimous among the incomparable and 268 of those 268 carrying task reversals (max 0.0796, median 0.0183), zero reversals among comparable pairs	measured in model, exploratory
CR-3 two entropy accountings	FAIL	closed forms to 8.9×10^{-16} , single-guess identity to 1.4×10^{-17} with 8 of 10 sources bit exact, Kraft equality exact and Huffman margins 0.0 and 0.0807 bits, ordering and equality clauses held exactly, but the declared clause requiring the difference of the two accountings to increase along the ladder measured 0, 0.398, 1.337, 2.275, 2.635, 2.113, 0.599 bits and turned over, because both accountings vanish together as the source becomes certain	measured in model, exploratory
CR-3b corrected declaration	PASS	the unchanged items repeated their values, the ratio falls strictly along both ladders, 1.0 to 0.1464 and 0.2876 to 0.1112, the closed-form derivative changes sign exactly once inside the declared bracket and agrees with a central finite difference to 1.6×10^{-8} , bisection in 39 steps to width 5.5×10^{-13} places the interior maximum at spike weight 0.35338 with separation 2.6398 bits, and the declared key-material source is charged 3.477 bits and exactly 1 bit by the two consumers	measured in model, exploratory
Scope	non-claim	the models here are exact finite toy models, this paper says nothing about computational hardness or asymptotic security, no real cipher, protocol, implementation, or deployed system is modeled or evaluated, no attack is developed, and every result is about evaluation methodology in declared finite models	non-claim

comparison in a declared finite model is a comparison for every task.

VII. METHODS AND REPRODUCIBILITY

All runs follow the campaign’s standing discipline. Substrates, channels, tasks, ladders, batteries, seeds, tolerances, and falsification bars were declared in the track document before the runs they govern, per the anti-circularity

contract, and verdicts are computed from the declared items rather than written first. One bar was moved before its run, when the object it had been drafted against was shown to be incapable of failing it, and the move, the reasoning, and a prediction for the abandoned object were written into the track document before the run so that the run could refute the prediction. Recording a pre-run deviation is not the same as applying one silently, and the record carries both the moved bar and the abandoned object’s measured values.

Every result-bearing run writes an append-only JSON evidence record containing the declared parameters, the measured outcomes, the computed verdict, and a canonical SHA-256 hash of the record body. The records for this paper are `cr0-instrument.json`, `cr1-leakage-ordering.json`, `cr2-countermeasure-flip.json`, `cr3-task-entropy.json`, and `cr3b-task-entropy.json` in the campaign repository, with the committed runners `cr0_instrument.py`, `cr1_leakage_ordering.py`, `cr2_countermeasure_flip.py`, `cr3_task_entropy.py`, and `cr3b_task_entropy.py` alongside them. The failed CR-3 record is retained unchanged and the corrected declaration is a separate record, so the sequence of verdicts is itself part of the evidence. A reducer may summarize records but may not rewrite them.

Every number and every figure coordinate in this paper is read from those committed records. The figure tables are generated deterministically by a committed

script, `make_figdata_cr.py`, which uses only the Python standard library and has no free parameters, and which asserts each value typed in this paper against the record value it came from before it writes anything, so a record that changed would break the build rather than disagree quietly with the text. The comparison curve of Fig. 1 is carried in the CR-0 record itself and is checked against the quantum track’s own record by the same script. The declared channel matrices, ladders, and costs quoted in the text are checked against the track document that declared them.

The CR-0 substrate was enumerated exactly with integer counting over all 29791 outcomes and the CR-3 and CR-3b substrates were built from exact rationals, so in those runs the only floating-point operations are the logarithms and the compensated sums that turn exact counts and exact rationals into bits. The CR-1 and CR-2 runs compute exact finite sums over declared joint distributions. All five runs executed on a single workstation in seconds and require no special hardware.

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